Homework 3.1: Bakeries, Cafés, and the Difference Between Monge and Kantorovich

Introduction

This homework studies discrete optimal transport using a bakery-café model.

Bakeries represent sources of bread supply, cafés represent destinations with bread demand, and transportation has a cost determined by the distance between locations.

The main objective is to compare two formulations of optimal transport:

- Monge transport, where each bakery must send all of its supply to exactly one café;
- Kantorovich transport, where a bakery may divide its supply among multiple cafés.

An LLM was used to help generate the examples, formulate the optimization problems, and generate Python code. The mathematical claims and computational results were then independently checked.

Part I: Generated Bakery and Café Data

LLM Prompt

I asked the LLM:

> Generate a small discrete optimal transport example representing bakeries and cafés in a two-dimensional city. Use 3 bakeries and 4 cafés, assign positive supplies and demands, and make sure total supply equals total demand. Choose simple coordinates so that Euclidean transportation costs can be calculated and visualized easily.

The generated problem contains

$$N = 3$$

bakeries and

$$M = 4$$

cafés.

Thus,

$$N < M.$$

Bakery Data

The bakery locations are

$$x_1 = (1, 1),$$

$$x_2 = (7, 1),$$

and

$$x_3 = (4, 7).$$

Their supplies are

$$a_1 = 5, \qquad a_2 = 6, \qquad a_3 = 4.$$

Therefore, total supply is

$$\sum_{i=1}^{3} a_i = 5 + 6 + 4 = 15.$$

Café Data

The café locations are

$$y_1 = (2, 3),$$

$$y_2 = (6, 3),$$

$$y_3 = (3, 6),$$

and

$$y_4 = (7, 6).$$

Their demands are

$$b_1 = 3, \qquad b_2 = 4, \qquad b_3 = 5, \qquad b_4 = 3.$$

Therefore, total demand is

$$\sum_{j=1}^{4} b_j = 3 + 4 + 5 + 3 = 15.$$

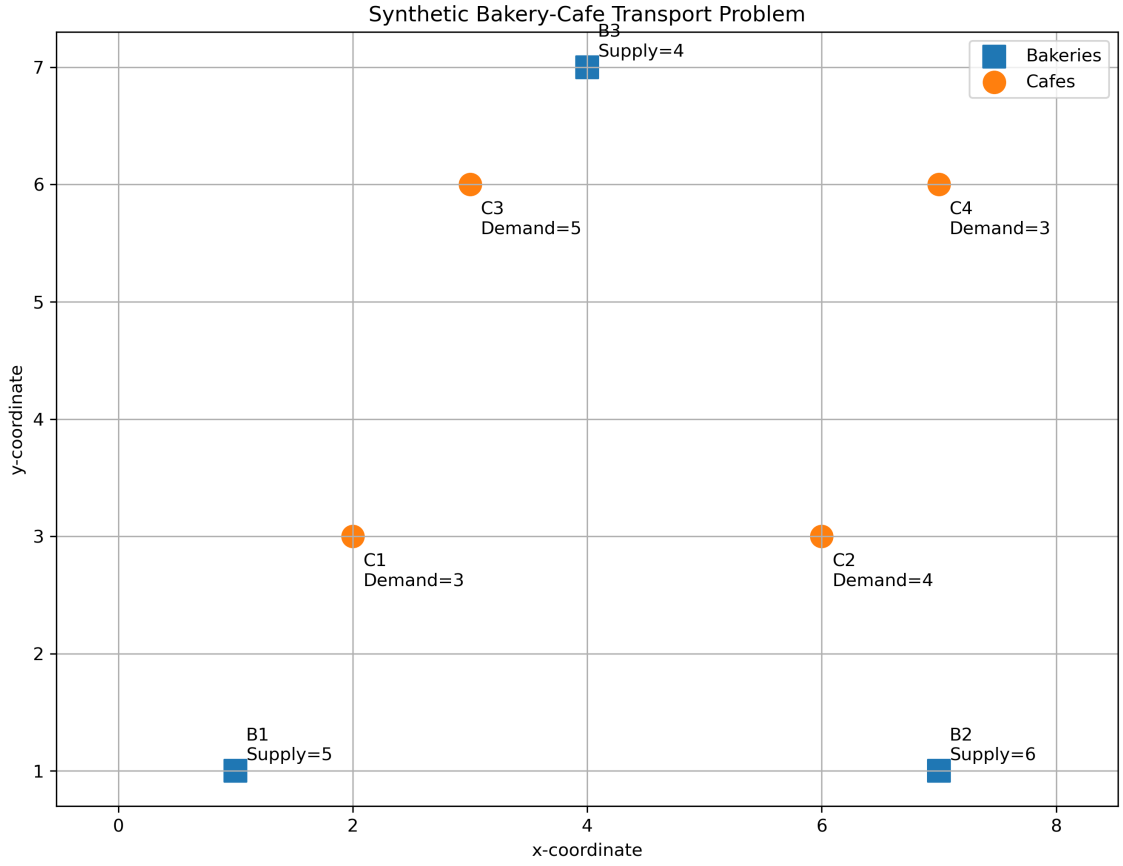
Thus,

$$\sum_{i=1}^{3} a_i = \sum_{j=1}^{4} b_j.$$

The transport problem is balanced.

Data Summary

Location	Coordinates	Supply or Demand
Bakery 1	$(1, 1)$	Supply 5
Bakery 2	$(7, 1)$	Supply 6
Bakery 3	$(4, 7)$	Supply 4
Café 1	$(2, 3)$	Demand 3
Café 2	$(6, 3)$	Demand 4
Café 3	$(3, 6)$	Demand 5
Café 4	$(7, 6)$	Demand 3



The Python verification confirmed that:

- all supplies are positive;
- all demands are positive;

- all locations lie in $\mathbb{R}^2$;
- total supply equals total demand.

Transportation Cost Matrix

The first transportation cost is Euclidean distance:

$$c_{ij} = |x_i - y_j|.$$

For points

$$x_i = (x_i^{(1)}, x_i^{(2)})$$

and

$$y_j = (y_j^{(1)}, y_j^{(2)}),$$

this is

$$c_{ij} = \sqrt{(x_i^{(1)} - y_j^{(1)})^2 + (x_i^{(2)} - y_j^{(2)})^2}.$$

The Euclidean cost matrix is

$$C = \begin{pmatrix} 2.2361 & 5.3852 & 5.3852 & 7.8102 \\ 5.3852 & 2.2361 & 6.4031 & 5.0000 \\ 4.4721 & 4.4721 & 1.4142 & 3.1623 \end{pmatrix}.$$

For example,

$$c_{11} = |(1,1) - (2,3)|.$$

Therefore,

$$c_{11} = \sqrt{(1-2)^2 + (1-3)^2} = \sqrt{5} \approx 2.2361.$$

The Python program independently calculated

$$c_{11} = 2.2360679775,$$

which exactly matched the manual calculation.

Squared Euclidean Cost

I also considered

$$c_{ij}^{(2)} = |x_i - y_j|^2.$$

The squared-distance cost matrix is

$$C^{(2)} = \begin{pmatrix} 5 & 29 & 29 & 61 \\ 29 & 5 & 41 & 25 \\ 20 & 20 & 2 & 10 \end{pmatrix}.$$

The program verified that

$$c_{ij}^{(2)} = (c_{ij})^2$$

for every pair of bakery and café locations.

Part II: The Kantorovich Problem

In the Kantorovich formulation, define

$$\gamma_{ij} \geq 0$$

to be the amount of bread sent from Bakery i to Café j.

The transport matrix is

$$\gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \gamma_{24} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & \gamma_{34} \end{pmatrix}.$$

Supply Constraints

Each bakery must ship exactly all of its supply:

$$\sum_{j=1}^{4} \gamma_{ij} = a_i.$$

Therefore,

$$\gamma_{11} + \gamma_{12} + \gamma_{13} + \gamma_{14} = 5,$$

$$\gamma_{21} + \gamma_{22} + \gamma_{23} + \gamma_{24} = 6,$$

and

$$\gamma_{31} + \gamma_{32} + \gamma_{33} + \gamma_{34} = 4.$$

Demand Constraints

Each café must receive exactly its demand:

$$\sum_{i=1}^{3} \gamma_{ij} = b_j.$$

Therefore,

$$\gamma_{11} + \gamma_{21} + \gamma_{31} = 3,$$
$$\gamma_{12} + \gamma_{22} + \gamma_{32} = 4,$$
$$\gamma_{13} + \gamma_{23} + \gamma_{33} = 5,$$

and

$$\gamma_{14} + \gamma_{24} + \gamma_{34} = 3.$$

Objective

The Kantorovich problem is

$$\min_{\gamma_{ij} \geq 0} \sum_{i=1}^{3} \sum_{j=1}^{4} c_{ij} \gamma_{ij}.$$

This is a linear programming problem.

I solved it using `scipy.optimize.linprog`.

Optimal Kantorovich Solution

For Euclidean cost, the optimal transport matrix was

$$\boxed{\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}}.$$

This means:

- Bakery 1 sends 3 units to Café 1 and 2 units to Café 3;

- Bakery 2 sends 4 units to Café 2 and 2 units to Café 4;
- Bakery 3 sends 3 units to Café 3 and 1 unit to Café 4.

Verification of Supply Constraints

The row sums are

$$3 + 2 = 5,$$
$$4 + 2 = 6,$$

and

$$3 + 1 = 4.$$

Thus,

$$(5, 6, 4)$$

is recovered exactly.

Verification of Demand Constraints

The column sums are

$$3,$$
$$4,$$
$$2 + 3 = 5,$$

and

$$2 + 1 = 3.$$

Thus,

$$(3, 4, 5, 3)$$

is also recovered exactly.

The program additionally verified nonnegativity and independently recomputed the objective value.

Optimal Euclidean Transportation Cost

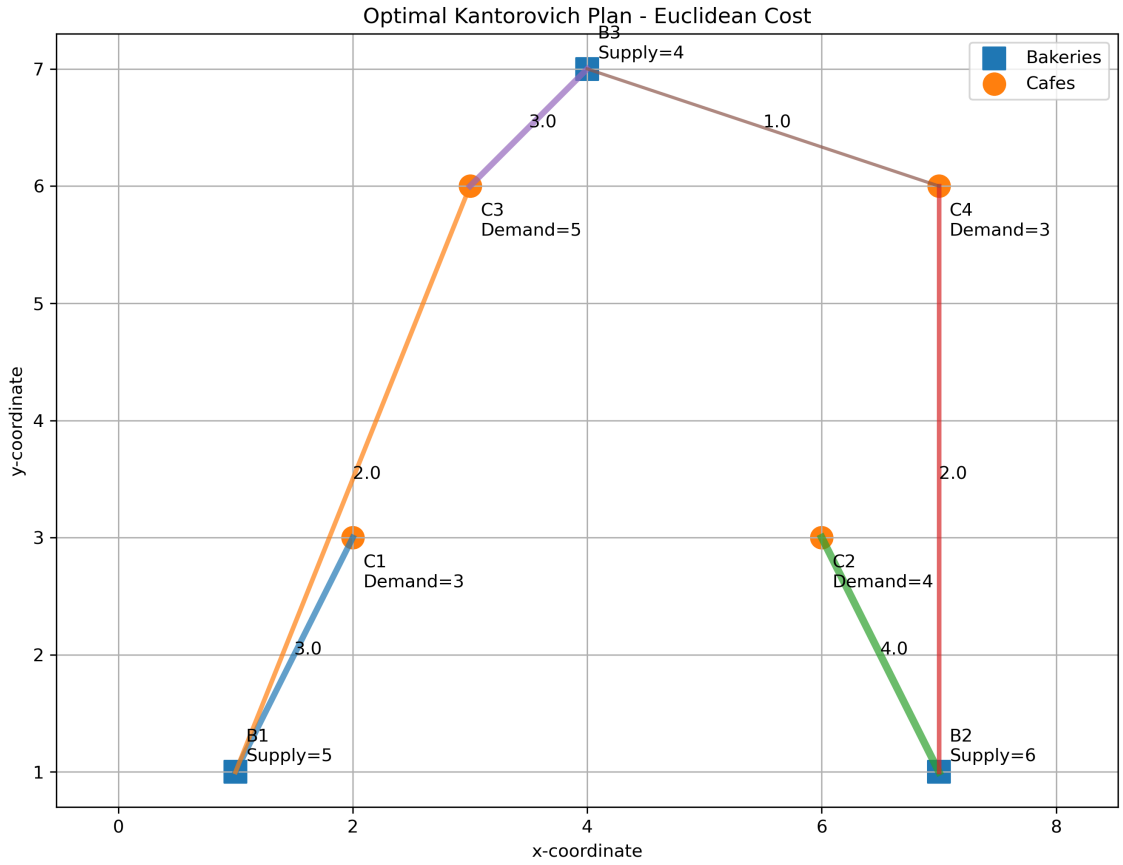
Only six entries of γ^* are nonzero.

Therefore,

$$J_K = 3\sqrt{5} + 2\sqrt{29} + 4\sqrt{5} + 2(5) + 3\sqrt{2} + \sqrt{10}.$$

Numerically,

$$J_K \approx 43.827724.$$



Squared Euclidean Kantorovich Solution

The squared-distance optimization produced the same transport matrix:

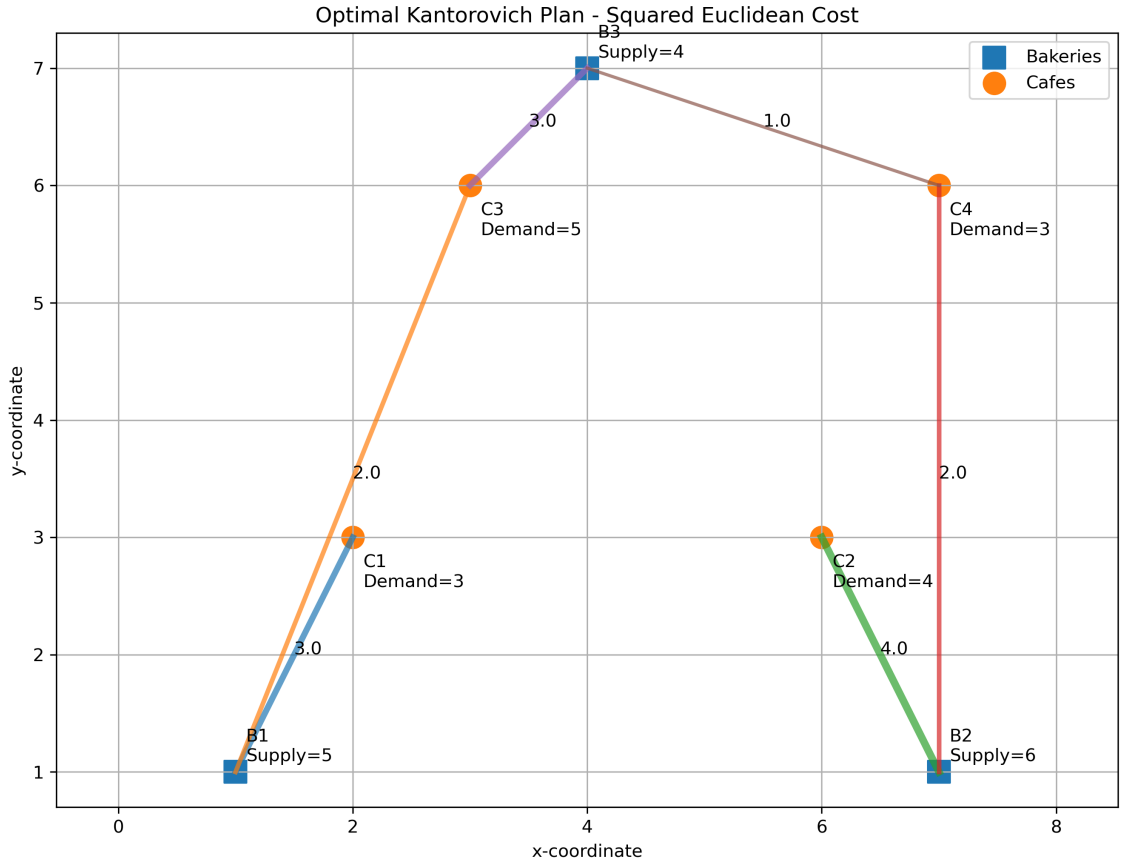
$$\gamma^{*(2)} = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Its cost is

$$3(5) + 2(29) + 4(5) + 2(25) + 3(2) + 1(10).$$

Therefore,

$$\boxed{J_K^{(2)} = 159}.$$



Thus both cost functions happened to produce the same optimal plan in this example, although this should not be expected for every transport problem.

Part III: The Monge Problem

In the Monge formulation, every bakery must send all of its supply to exactly one café.

A Monge map has the form

$$T : \{1, 2, 3\} \to \{1, 2, 3, 4\}.$$

The café constraints require

$$\sum_{i:T(i)=j} a_i = b_j.$$

The Monge cost would be

$$J_M(T) = \sum_{i=1}^{3} a_i c_{i,T(i)}.$$

Number of Possible Maps

Each of the three bakeries has four possible destinations.

Therefore,

$$4^3 = 64$$

Monge maps are possible.

The Python program exhaustively searched all 64 maps.

It found

$$0$$

admissible Monge maps.

Thus,

$$\boxed{\text{the Monge problem is infeasible for this city.}}$$

Why No Monge Map Exists

There are

$$N = 3$$

bakeries but

$$M = 4$$

cafés.

Every café has positive demand.

Because one bakery can serve only one café under Monge transport, three bakeries can serve at most three distinct cafés.

But four cafés require positive mass.

Therefore, at least one café would necessarily receive zero bread.

Thus no Monge map can satisfy the demand constraints.

There is also a second obstruction.

The possible sums formed from the indivisible bakery supplies

$$5,\ 6,\ 4$$

are

$$0,\ 4,\ 5,\ 6,\ 9,\ 10,\ 11,\ 15.$$

However, two cafés require only

$$3$$

units.

No subset of whole bakery supplies sums to 3.

Thus the specific supply and demand values also prevent a Monge solution.

Comparison with Kantorovich

The Kantorovich solution was

$$\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Each bakery splits its supply.

For example,

$$5 = 3 + 2,$$

$$6 = 4 + 2,$$

and

$$4 = 3 + 1.$$

This splitting is exactly what allows Kantorovich transport to satisfy all four cafés.

Therefore,

$$\boxed{\text{Kantorovich is feasible}}$$

while

$$\boxed{\text{Monge is infeasible}}.$$

Part IV: Comparison of $N > M$ and $N < M$

Two additional cases were compared.

Case 1: More Bakeries Than Cafés

For Case 1,

$$N = 4, \qquad M = 3.$$

The supplies are

$$a = (2, 2, 2, 2),$$

and demands are

$$b = (4, 2, 2).$$

The bakery locations are

$$(0, 0), \quad (0, 6), \quad (8, 0), \quad (8, 6),$$

and the café locations are

$$(0, 3), \quad (7, 0), \quad (7, 6).$$

The Euclidean cost matrix is

$$C \approx \begin{pmatrix} 3.0000 & 7.0000 & 9.2195 \\ 3.0000 & 9.2195 & 7.0000 \\ 8.5440 & 1.0000 & 6.0828 \\ 8.5440 & 6.0828 & 1.0000 \end{pmatrix}.$$

Case 1 Monge Solution

There are

$$3^4 = 81$$

possible Monge maps.

The exhaustive search found

$$12$$

feasible maps.

The optimal map was

$$T(1) = 1, \qquad T(2) = 1, \qquad T(3) = 2, \qquad T(4) = 3.$$

Its cost is

$$J_M = 2(3) + 2(3) + 2(1) + 2(1).$$

Thus,

$$J_M^* = 16.$$

Case 1 Kantorovich Solution

The optimal Kantorovich matrix was

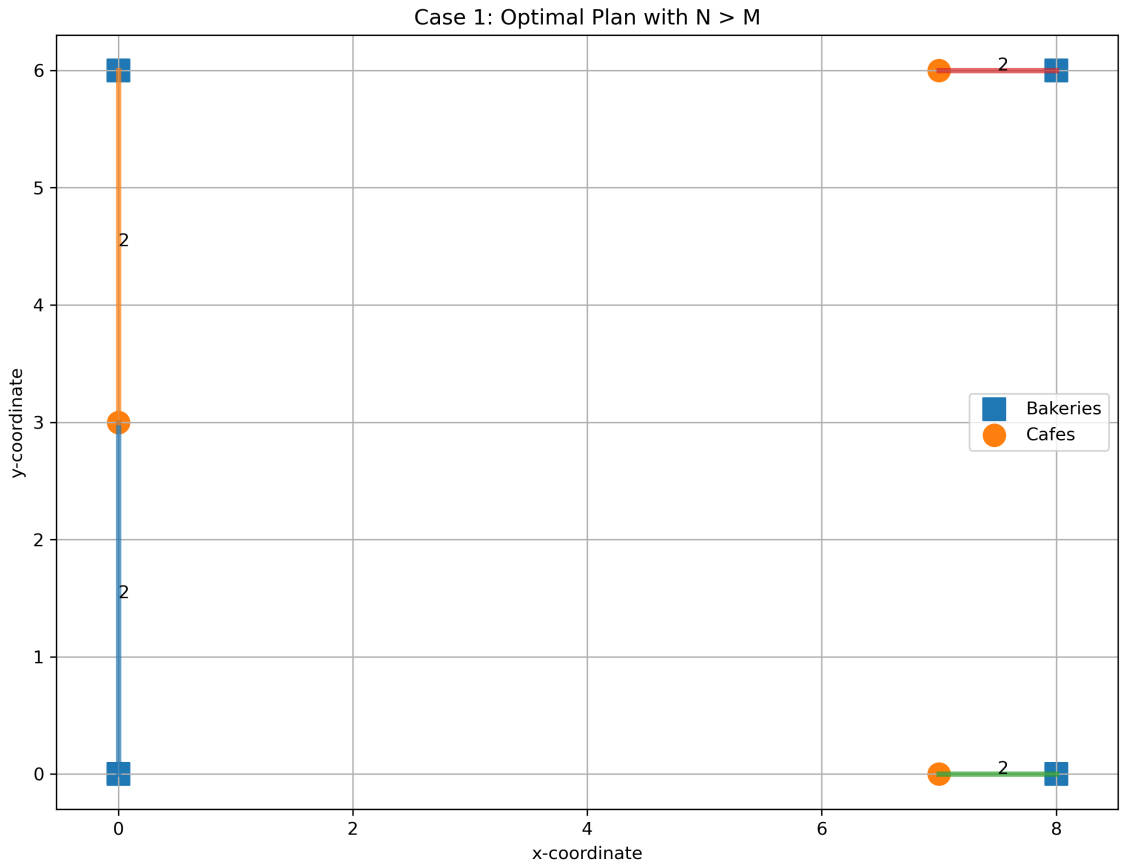
$$\gamma^* = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Its cost was also

$$J_K^* = 16.$$

Each bakery has exactly one positive destination.

Therefore, no mass splitting is required, and the displayed optimal Kantorovich plan is induced by the optimal Monge map.



Thus,

$$J_M^* = J_K^* = 16.$$

Case 2: Fewer Bakeries Than Cafés

For Case 2, the original city was reused:

$$N = 3, \qquad M = 4.$$

The exhaustive Monge search found

$$0$$

feasible maps.

Thus,

$$\boxed{\text{Monge is infeasible.}}$$

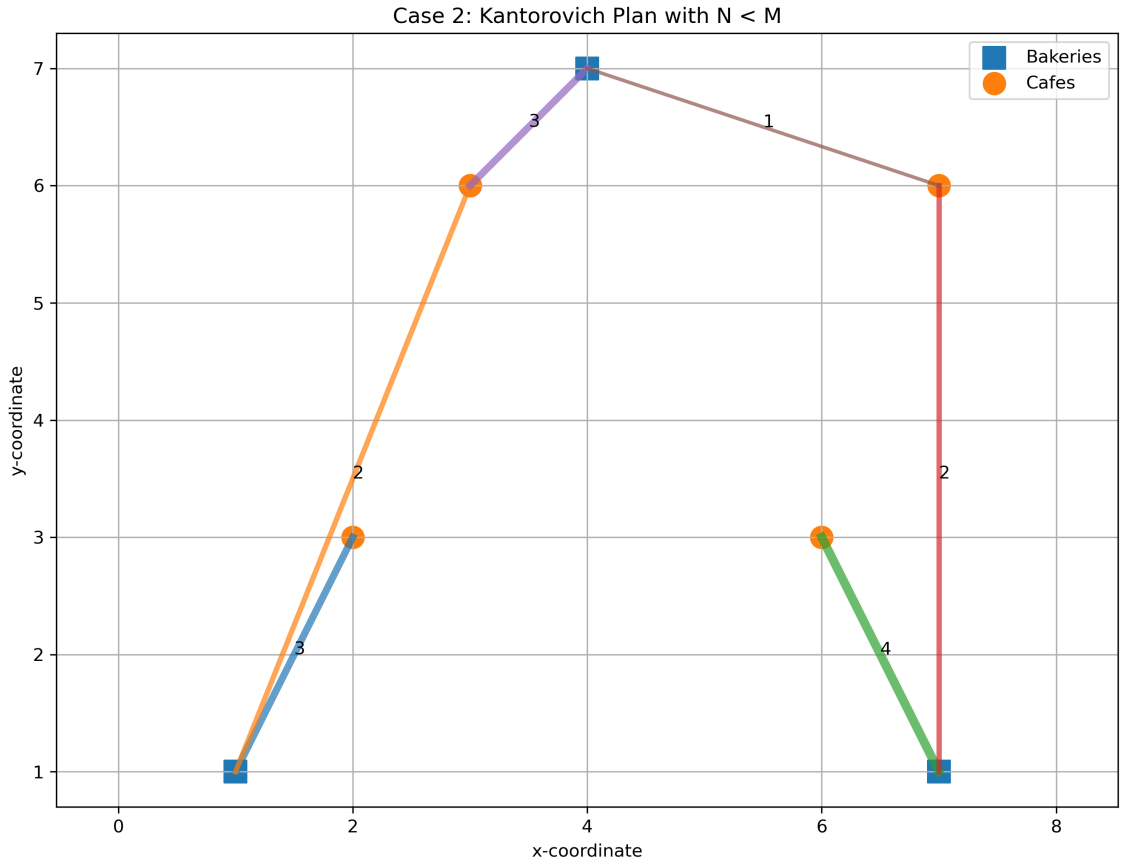
The Kantorovich problem remained feasible and produced

$$\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Its Euclidean cost was

$$J_K^* \approx 43.827724.$$

Every bakery had two positive destinations, so every bakery split its supply.



Summary of the Two Cases

Property	$N > M$	**Case 1:**	**Case 2:** $N < M$
Number of bakeries		4	3
Number of cafés		3	4

Property	Case 1: $N > M$	Case 2: $N < M$
Possible Monge maps	81	64
Feasible Monge maps	12	0
Monge feasible?	Yes	No
Optimal Monge cost	16	Does not exist
Kantorovich feasible?	Yes	Yes
Optimal Kantorovich cost	16	43.827724
Kantorovich requires splitting?	No	Yes

The experiments show that

$$N > M$$

does not automatically guarantee a Monge solution.

The supplies must also be compatible with the café demands.

In Case 1,

$$4 = 2 + 2,$$

so the café demands can be constructed from whole bakery supplies.

By contrast, if

$$N < M$$

and every café has positive demand, then a discrete Monge map is impossible because each bakery has only one destination.

Thus,

$$N < M \text{ and all } b_j > 0 \Longrightarrow \text{no Monge map exists.}$$

Kantorovich avoids this obstruction by permitting mass splitting.

Part V: LLM Prompts and Critique

Prompt for Generating the Synthetic City

> Generate a small discrete optimal transport example representing bakeries and cafés in a two-dimensional city. Use 3 bakeries and 4 cafés, assign positive supplies and demands, and make sure total supply equals total demand. Choose simple coordinates so that Euclidean transportation costs can be calculated and visualized easily.

Prompt for the Monge and Kantorovich Critique

> Explain the difference between Monge and Kantorovich optimal transport using a discrete bakery-café problem. Provide the mathematical formulations for both approaches and generate Python code to solve them using the same transportation example.

The second LLM generated mathematical formulations and Python code using:

- `scipy.optimize.linprog` for Kantorovich transport;
- `scipy.optimize.milp` for Monge transport.

I checked the mathematics and code carefully.

---

1. Supply Constraints

For Kantorovich transport, the LLM correctly imposed

$$\sum_j x_{ij} = a_i.$$

Its code constructs one equality constraint for each bakery.

For Monge transport, it introduces binary variables

$$z_{ij} \in \{0, 1\},$$

with

$$\sum_j z_{ij} = 1.$$

The resulting shipment is

$$x_{ij} = a_i z_{ij}.$$

Thus every bakery sends all of its supply to exactly one café.

The supply constraints are correct.

---

2. Demand Constraints

For Kantorovich transport, the LLM correctly imposed

$$\sum_i x_{ij} = b_j.$$

For Monge transport, it correctly imposed

$$\sum_i a_i z_{ij} = b_j.$$

Thus the demand constraints are also mathematically correct.

There was one notation typo in the written explanation.

It wrote

$$\sum_{i(i)=j} a_i = b_j,$$

when it should have written

$$\sum_{i:T(i)=j} a_i = b_j.$$

The implementation itself used the correct condition.

---

3. Transport Map Versus Transport Plan

The LLM correctly distinguished a Monge transport map from a Kantorovich transport plan.

Monge uses

$$T(i) = j,$$

meaning that the entire source mass a_i is assigned to one destination.

Kantorovich instead uses the matrix

$$(x_{ij}),$$

and multiple entries in one row may be positive.

Every feasible Monge map induces a Kantorovich plan through

$$x_{ij} = \begin{cases} a_i, & j = T(i), \\ 0, & j \neq T(i). \end{cases}$$

However, not every Kantorovich plan comes from a Monge map.

Thus,

$$\{\text{Monge-induced plans}\} \subseteq \{\text{Kantorovich plans}\}.$$

The LLM handled this distinction correctly.

---

4. Does a Monge Map Always Exist?

The LLM correctly stated that equal total supply and demand do not guarantee a Monge solution.

Its second example used

$$a = (5, 5)$$

and

$$b = (6, 4).$$

Total supply and demand are both

$$10,$$

but no bakery can split its supply.

Therefore, the demands 6 and 4 cannot be formed from two indivisible masses of size 5.

The LLM correctly reported that the Monge problem is infeasible.

Kantorovich can instead use

$$X = \begin{pmatrix} 5 & 0 \\ 1 & 4 \end{pmatrix},$$

which has the correct row and column sums.

The transportation cost is

$$5(2) + 1(3) + 4(4) = 29.$$

Thus this example is mathematically correct.

---

5. Handling the Case $N < M$

This was the main weakness of the LLM response.

The generated solver is general enough to handle

$$N < M.$$

However, neither original example actually tested this case.

Example 1 had

$$N = 3 > M = 2,$$

while Example 2 had

$$N = M = 2.$$

Therefore, the second example showed Monge infeasibility but did not directly test the structural case required in the homework.

I added an additional test using the original synthetic city:

$$N = 3 < M = 4.$$

The supplies were

$$(5, 6, 4),$$

and demands were

$$(3, 4, 5, 3).$$

The same LLM-generated solver returned the Kantorovich plan

$$\begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}$$

with cost

$$43.827724.$$

The Monge solver correctly reported that no feasible Monge solution exists.

Thus the generated code handles $N < M$ correctly even though the LLM did not initially include an example testing it.

Verification of the LLM's First Example

The LLM's first example used

$$a = (2, 3, 5),$$

$$b = (5, 5),$$

and

$$C = \begin{pmatrix} 2 & 6 \\ 3 & 4 \\ 6 & 1 \end{pmatrix}.$$

The Monge assignment

$$T(1) = 1, \qquad T(2) = 1, \qquad T(3) = 2$$

induces

$$X = \begin{pmatrix} 2 & 0 \\ 3 & 0 \\ 0 & 5 \end{pmatrix}.$$

Its cost is

$$2(2) + 3(3) + 5(1) = 18.$$

Both the Monge and Kantorovich optimizers give

$$18.$$

Thus the LLM's first example is correct.

Overall LLM Reliability

The LLM performed well on the central mathematics.

It correctly:

- formulated the Kantorovich supply constraints;
- formulated the Kantorovich demand constraints;

- implemented the Monge one-destination constraint;
- implemented the Monge demand constraints;
- distinguished a transport map from a transport plan;
- recognized that Monge feasibility is not guaranteed;
- recognized Kantorovich as a relaxation of Monge;
- correctly concluded that

$$C_K^* \leq C_M^*$$

whenever the Monge problem is feasible.

The main weakness was completeness rather than mathematical correctness.

The LLM was explicitly supposed to handle the case

$$N < M,$$

but its original examples did not test that case.

After adding the missing test, its code produced the mathematically correct result.

Thus the LLM was useful for generating the formulation and code, but its response still required independent checking to ensure that all requested cases had actually been tested.

Final Conclusion

The experiments demonstrate the central difference between Monge and Kantorovich optimal transport.

Monge transport requires a deterministic map:

$$T : \{1, \ldots, N\} \to \{1, \ldots, M\},$$

where each bakery sends its entire mass to one café.

Kantorovich transport instead allows a transport matrix

$$\gamma = (\gamma_{ij}),$$

where mass may be divided among multiple destinations.

Every Monge solution induces a Kantorovich plan, so whenever Monge is feasible,

$$C_K^* \leq C_M^*.$$

In the $N > M$ example, a compatible Monge map existed and both formulations had the

same optimal cost:

$$C_M^* = C_K^* = 16.$$

In the $N < M$ example, no Monge map existed, while Kantorovich remained feasible through mass splitting:

$$C_K^* \approx 43.827724.$$

Therefore, Kantorovich transport is more general than Monge transport.

The computational experiments also show why LLM-generated mathematics should be verified. The LLM's formulation and code were largely correct, but it initially failed to demonstrate one of the specific cases requested by the assignment.